

CY 562: Stereochemistry of Organic Molecules

Course Code: CY 562

Course Name: Stereochemistry of Organic Molecules

L-T-P-C: 3-0-0-3

Prerequisites: None

Intended for: UG/PG/PhD

Mutual Exclusion: NA

Distribution: Elective

Approval: 65th BoA

Preamble

This course aims to provide a comprehensive understanding of fundamental and advanced concepts in the stereochemistry of organic molecules, including molecular symmetry, chirality, and conformational analysis. The course also covers the stereochemical aspects of organic reactions, including addition, substitution, and elimination reactions, along with the applications of stereochemistry in modern organic synthesis and biological systems. Students will gain in-depth knowledge of stereochemical principles, molecular structure, reactivity, and their significance in contemporary chemical research.

Course Modules

- **Module 1: Fundamentals of Stereochemistry:** Molecular symmetry and its relationship with chirality; symmetry elements and operations (plane of symmetry, centre of symmetry and improper rotation axis); chiral and achiral molecules; stereogenic centres; enantiomers, diastereomers, meso compounds and racemic mixtures. Optical activity, specific rotation, optical purity and resolution of racemic mixtures. Representation of stereochemical structures using Fischer, Newman and Sawhorse projections. Relative and absolute configuration; Cahn–Ingold–Prelog (CIP) sequence rules; assignment of R/S and E/Z configurations. (7 Hours)
- **Module 2: Conformational Analysis of Organic Molecules:** Conformational analysis of acyclic systems including ethane, substituted alkanes and factors affecting conformational stability such as torsional strain, steric interactions, gauche effect and allylic strain ($A^{1,3}$ strain). Conformations of cyclic systems including cyclopropane, cyclobutane, cyclopentane and cyclohexane; chair, boat and twist-boat conformations, ring inversion, axial and equatorial orientations, and conformational preferences of substituted cyclohexanes. Conformational analysis of bicyclic, tricyclic, fused, bridged and spirocyclic systems, including cis/trans-decalin. Conformations of saturated heterocyclic systems and the role of stereoelectronic effects, including the anomeric effect. Dynamic conformational processes such as ring inversion, nitrogen inversion and conformational effects on molecular stability and reactivity. (10 Hours)
- **Module 3: Stereochemistry of Organic Reactions:** Stereochemical aspects of organic reactions including nucleophilic substitution (S_N1 and S_N2), elimination ($E1$ and $E2$), and electrophilic and nucleophilic addition reactions. Concepts of stereospecific and stereoselective reactions, diastereoselectivity and enantioselectivity. Prochirality, topicity (homotopic, enantiotopic, and diastereotopic groups), Re and Si face selectivity, and facial selectivity. Neighboring group participation and stereochemical control in organic reaction mechanisms. (10 Hours)
- **Module 4: Advanced Stereochemistry and Applications:** Advanced concepts of chirality including central chirality beyond carbon (chirality at phosphorus, sulfur, silicon, nitrogen, etc.), axial chirality (al-

lenes, biphenyls and atropisomerism), planar chirality, helical chirality inherent chirality, and nitrogen stereogenicity (pyramidal nitrogen chirality). Dynamic stereochemistry including configurational stability, racemization and restricted rotation. Principles of asymmetric synthesis, asymmetric induction, chiral auxiliaries, chiral reagents and chiral catalysts. Applications of stereochemistry in natural products, carbohydrates, amino acids and medicinal chemistry. Determination of stereochemistry using modern analytical techniques such as NMR spectroscopy, circular dichroism (CD) and X-ray crystallography. (15 Hours)

Text Books

1. Eliel, E. L., Wilen, S. H. and Mander, L. N., *Stereochemistry of Organic Compounds*, Wiley, New York, 1994.
2. Nasipuri, D., *Stereochemistry of Organic Compounds: Principles and Applications*, 4th Edition, New Academic Science Limited, 2012.

References

1. Wu, S., Chen, P., Duan, M., Jiang, P.-Y., Zhou, Q., Xiang, S.-H., Houk, K. N. and Tan, B., *Controlling Pyramidal Nitrogen Chirality by Asymmetric Organocatalysis*, *Nature*, 2025, 647, 897–905.
2. Zhu, C., Das, S., Sterling, M. S., Tsuji, N., Léger, S. J., Neese, F., De, C. K. and List, B., *The Asymmetric Synthesis of an Acyclic N-Stereogenic Amine*, *Nature*, 2026, 649, 621–625.
3. Cheng, J. K., Xiang, S.-H., Li, S., Ye, L. and Tan, B., *Recent Advances in Catalytic Asymmetric Construction of Atropisomers*, *Chemical Reviews*, 2021, 121(8), 4805–4902.
4. Kalsi, P. S., *Stereochemistry: Conformation and Mechanism*, 10th Edition, New Age International Publishers, New Delhi, 2017.
5. Clayden, J., Greeves, N. and Warren, S., *Organic Chemistry*, 2nd Edition, Oxford University Press, Oxford, 2012.
6. Zweifel, G. S., Nantz, M. H. and Somfai, P., *Modern Organic Synthesis: An Introduction*, 2nd Edition, John Wiley & Sons, Hoboken, NJ, 2017.
7. Walsh, P. J., Kozlowski, M. C., *Fundamentals of Asymmetric Catalysis*, University Science Books, California, 2008.